

An AI-Powered Workforce Trend Analysis and Career Intelligence Platform

¹Vallapuneni Gopii Chennduu, ²Gavara Satya Srinivas, ³Thammana Venkata Rama Durga Naga Sairam, ⁴Prof. Ryagnirvesh Kammari

¹gopii.vallapaneni127095@marwadieducation.ac.in,

²gavarasatyasrinivas.127013@marwadiuniversity.ac.in,

³thammanavenkataramadurganagasairam.126512@marwadiuniversity.ac.in,

⁴ryagnirvesh.kammari@marwadieducation.edu.in

¹²³⁴Department of Computer Engineering- AI, ML & DS
Marwadi University, Rajkot, Gujarat, India

Abstract- I've never seen the job market this unstable before, at least not in my experience. It doesn't even feel like change is on the way anymore it's already here. And honestly, it's moving way faster than I thought it would. AI, cloud tech, automation. . . people used to talk about these like future things. Now you can actually see them affecting real jobs around you. Some companies that once felt completely secure don't look that strong anymore, at least not the way they used to. And at the same time, new ones just keep popping up out of nowhere and growing really fast. It's a bit unpredictable. Even the things that felt important to learn six months ago don't always feel that useful today. Jobs are just disappearing. . . sometimes you don't even see it coming until it's already happened. At the same time, new roles are coming up but it's not always clear how to move into them. Layoffs have become so common that people don't react the same way anymore, which feels a bit strange when you think about it. Because of all this, people keep second guessing themselves about what to do next. Like, even simple decisions don't really feel that simple anymore. And figuring out how to move forward how to actually stay relevant with everything changing so fast that's honestly the hard part.. Job boards and LinkedIn are still there, of course, but even they feel like they're struggling to keep tempo. But more often than not, they're little more than a job exchange. What is actually going on? This paper brings something different. The Career Platform with AI-Intelligence (CPI) is unique in that it is not just another platform, but an intelligent thinking system. Drawing data from job portals, technical trends, layoff news, and even the anonymous signals of employees themselves. It then presents things in an easily digestible form. Extracting skills through NLP, Role Stability Scores and skill gap highlights This entire process includes insights tailored to each user. It's a bit like a guide, but far more intelligent. The entire platform utilizes a multi-layer (front end running React, Google OAuth back end, Node.js), an AI engine strung together with Python, and a scalable ETL pipeline. It's all quite complex, of course. But this mess is actually made to be understood

Keywords: Career Platform, AI-intelligence, Workforce, Trend Analysis, Skill Demand Analytics, Role Stability, Layoff Awareness, SaaS Platform, NLP, ETL Pipeline, Data-Driven, Career Planning, Skill Gap Radar, Workforce Volatility Modeling, Role-Based, Dashboards, Career Stability, Career Insights, Job Market

References

- [1] S. Berke and A. Tick, “Managing Across Generations: Organizational Problems and Solutions in a Changing Workforce,” in Proc. IEEE SAMI 2026 (Int. Symp. Applied Machine Intelligence and Informatics), Stará Lesná, Slovakia, pp. 1–6, Jan. 2026, doi: 10.1109/SAMI68106.2026.11420675.
- [2] M. Ansell, N. Papakonstantinou, and D. Van Bossuyt, “Modeling Knowledge Retention in PRA Models for Mission Critical Infrastructure,” in Proc. IEEE RAMS 2026 (Annual Reliability and Maintainability Symp.), Miramar Beach, FL, USA, pp. 1–6, Jan. 2026, doi: 10.1109/RAMS50514.2026.11424434.
- [3] A. Dhivya, P. Ambika, and K. Thirumalvalavan, “Predictive Analytics for Enhancing Employee Retention through Strategic Career Development in IT Companies,” in Proc. IEEE ICEARS 2026 (Int. Conf. on Engineering and Applied Research Sciences), Tuticorin, India, pp. 1–6, Feb. 2026, doi: 10.1109/ICEARS67481.2026.11416675.
- [4] P. C. Saranya and K. S. Singh, “A Novel Hybrid Machine Learning Approach for Employee Future Prediction: Integrating Advanced Feature Engineering with Ensemble Methods,” in Proc. IEEE ICECONF 2025 (Int. Conf. on Engineering and Computing), Chennai, India, pp. 1–6, Oct. 2025, doi: 10.1109/ICECONF65644.2025.11379435.
- [5] G. N. Tóth and Á. Drégelyi-Kiss, “Bridging Skill Gaps through STEM: HR Risk Management in Higher Education for Industry 4.0 and 5.0,” in Proc. IEEE SAMI 2026 (Int. Symp. Applied Machine Intelligence and Informatics), Stará Lesná, Slovakia, pp. 1–6, Jan. 2026, doi: 10.1109/SAMI68106.2026.11420348.
- [6] S. R. Pyreddy, K. V. Pathan, H. Masum, and T. S. Zaman, “SECite: Analyzing and Summarizing Citations in Software Engineering Literature,” in Proc. IEEE CCWC 2026 (Computing and Communication Workshop and Conf.), Las Vegas, NV, USA, pp. 1–6, Jan. 2026, doi: 10.1109/CCWC67433.2026.11393720.
- [7] S. Huang and J. Hu, “Demand for Long-Term-Oriented Human Capital as a Strategic Response to Innovation Aspiration Gaps: An Analysis of Online Job Postings by Firms,” IEEE Trans. Eng. Manage., pp. 1–13, Feb. 2026, doi: 10.1109/TEM.2026.3667890.
- [8] L. Wang, “Deep Learning for College Graduates Employment Prediction: A Computational Approach,” IEEE Access, vol. 14, pp. 32273–32294, Feb. 2026, doi: 10.1109/ACCESS.2026.3661553.
- [9] J. Feng, Y. Zhao, X. Zhong, Q. Miao, X. Peng, and L. Zhang, “A Job-Seeking Assistance System by Integrating Interview Training With Job Recommendation,” IEEE Trans. Learn. Technol., vol. 19, pp. 51–62, Jan. 2026, doi: 10.1109/TLT.2026.3659507.
- [10] K. Gao, Y. Hu, C. Nie, and W. Li, “Deep Q-Learning-Based Intelligent Scheduling for ETL Optimization in Heterogeneous Data Environments,” in Proc. IEEE ISCEIC 2025 (Int. Symp. on Computer Engineering and Intelligent Communications), Chongqing, China, pp. 1–6, Nov. 2025, doi: 10.1109/ISCEIC67854.2025.11405485.
- [11] W. Kumwilaisak, “Intelligent EMS Dispatch via Adaptive Learning With Counterfactual and Risk-Aware Optimization,” IEEE Access, vol. 14, pp. 31804–31823, Feb. 2026, doi: 10.1109/ACCESS.2026.3668258.
- [12] M. Roy and K. Sharma, “AI-Powered Career Guidance: A Scalable Model for Personalized Recommendations,” IEEE Access, vol. 14, pp. 20927–20941, Feb. 2026, doi: 10.1109/ACCESS.2026.3668258.

10.1109/ACCESS.2026.3661573.

- [13] G. Rak, “Early-Career Engineers vs. AI: As AI Takes on More Work in 2026, Expectations for Entry-Level Roles Grow,” *IEEE Spectr.*, pp. 40–42, Jan. 2026, doi: 10.1109/MSPEC.2026.11345562.
- [14] I. Pardi and A. Bánáti, “Transitioning from Windows-based to Linux-based Cyber Ranges: Design, Automation, and Educational Impact,” in *Proc. IEEE SAMI 2026 (Int. Symp. Applied Machine Intelligence and Informatics)*, Stará Lesná, Slovakia, pp. 1–6, Jan. 2026, doi: 10.1109/SAMI68106.2026.11420822.
- [15] R. K. Kanji, N. S. H. Rellu, V. R. Surasani, and V. R. Mekala, “Evaluating the Impact of AI on Data Quality and Governance in Cloud-based Systems,” in *Proc. IEEE ICEARS 2026 (Int. Conf. on Engineering and Applied Research Sciences)*, Tuticorin, India, pp. 1–6, Feb. 2026, doi: 10.1109/ICEARS67481.2026.11416540.
- [16] S. M. Khaniabadi, S. Soares, and S. I. Lopes, “Edge-AI Technologies for Real-Time Industrial Object Identification and Tracking: An Overview,” in *Proc. IEEE CiSt 2025 (Congress on Information Science and Technology)*, Marrakech, Morocco, pp. 1–6, Oct. 2025, doi: 10.1109/CiSt65886.2025.11224223.
- [17] R. R. Skandarsini, T. Lakshmi Narayanaa, and S. J. Ida, “Forecasting Unemployment Recession Rates: A Logistic Regression Approach Utilizing Comprehensive Economic Indicators,” in *Proc. IEEE RMKMATE 2023 (Int. Conf. on Recent Advances in Materials and Technology)*, Chennai, India, pp. 1–6, Nov. 2023, doi: 10.1109/RMK-MATE59243.2023.10368645.
- [18] L. R. Pelima, Y. Sukmana, and Y. Rosmansyah, “Predicting University Student Graduation Using Academic Performance and Machine Learning: A Systematic Literature Review,” *IEEE Access*, vol. 12, pp. 23451–23465, Feb. 2024, doi: 10.1109/ACCESS.2024.3361479.
- [19] P7015/D2, *IEEE Draft Standard for Data and Artificial Intelligence (AI) Literacy, Skills, and Readiness*, IEEE, Mar. 2026. [Online]. Available: <https://ieeexplore.ieee.org/servlet/opac?punumber=11429665>